Polynomial parametrization of a 3-by-3 magic square of perfect squares

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Abstract

This paper aims at proposing a refined method to search for a magic square of perfect squares of size 3 with distinct entries, also known as "Parker square". After defining some algebra to potentially parametrize the problem, a more advanced way involving polynomials is described, together with a summary of the algorithm used in this work. Despite no actual solution was found in this scope, a close approximation is exhibited.

1 Definition of the problem

Consider the following structure:

$$\begin{matrix} A^2 & B^2 & C^2 \\ D^2 & E^2 & F^2 \\ G^2 & H^2 & I^2 \end{matrix} \tag{1}$$

with $A, B, C, D, E, F, G, H, I$ all distinct natural numbers. It is a magic square when the sum of every line, every column and both diagonals equals to the same value. In mathematical language this translates to: $\exists S \in \mathbb{N}$ such that all the following equations are satisfied:

$$\begin{cases} A^2 + B^2 + C^2 = S \\ D^2 + E^2 + F^2 = S \\ G^2 + H^2 + I^2 = S \\ A^2 + D^2 + G^2 = S \\ B^2 + E^2 + H^2 = S \\ C^2 + F^2 + I^2 = S \\ A^2 + E^2 + I^2 = S \\ G^2 + E^2 + C^2 = S \end{cases} \tag{2}$$

It is a long term recreational mathematics problem [1] and as of today the existence of a solution is unknown [4, 2]; it could neither be found or proven impossible. Following the set of equations 2 it is easy to rewrite the structure 1 with two integer parameters $n = A^2 - E^2$ and $m = C^2 - E^2$ as follows:

$$\begin{array}{ccc} E^2+n & E^2-n-m & E^2+m \\ E^2-n+m & E^2 & E^2+n-m \\ E^2-m & E^2+n+m & E^2-n \end{array} \tag{3}$$

Given the symmetry of structure 3 the search will be restricted to $n, m > 0$ and furthermore to $n > m$ only. In order to better visualize the method used in this paper, it is temporarily assumed that $n > 2m$; in that case the order of the elements is shown here, together with the distance between them (written above the inequality sign):

$$B^2 \overset{m}{<} I^2 \overset{m}{<} D^2 \overset{n-2m}{<} G^2 \overset{m}{<} E^2 \overset{m}{<} C^2 \overset{n-2m}{<} F^2 \overset{m}{<} A^2 \overset{m}{<} H^2 \tag{4}$$

What can be deduced from above is that both n and m are triple congrua; this is also valid when dropping the assumption $n > 2m$, which will be done from here. For instance, writing $m = C^2 - E^2 = E^2 - G^2$ means that m is a congruum with the arithmetic progression of the squares of $\{G, E, C\}$. It is also with $\{B, I, D\}$ in a lower range for all elements and $\{F, A, H\}$ in a higher range. Similarly, n is a triple congruum with $\{B, G, F\}$, $\{I, E, A\}$ and $\{D, C, H\}$, in this order. The search will focus on one of those, namely $\{I, E, A\}$; both remaining conditions on arithmetic progressions will be fulfilled in that case, because of conditions on m.

2 Congruum parametrization

As a congruum, we can write that $m = 4ab\left(b^2 - a^2\right)$ with a and b distinct natural numbers. The middle number to be squared is a^2+b^2 and the others $b^2 - a^2 \pm 2ab$. Hence, as a triple congruum, two additional pairs are satisfying $m = 4cd\left(d^2 - c^2\right)$ and $m = 4ef\left(f^2 - e^2\right)$. In other words, finding m is equivalent to finding (a, b, c, d, e, f) with $a < c < e$ and such that:

$$ab\left(b^2 - a^2\right) = cd\left(d^2 - c^2\right) = ef\left(f^2 - e^2\right) \tag{5}$$

An example of set fulfilling that property is given by $(a, b, c, d, e, f) = (3, 7, 5, 7, 7, 8)$ leading to $m = 3360$. Another example that is in the same order as inequation 4 is $(a, b, c, d, e, f) = (7, 37, 33, 37, 37, 40)$ with $m = 1367520$. There seems to be an infinite number of these sets (where all numbers are relatively prime). Given equation 5 is satisfied, n is well defined (i.e. the initial problem has a solution) if, for example, the middle squares of each arithmetic progression for m are forming themselves an arithmetic progression, which can be written as $\left(c^2+d^2\right)^2 - \left(a^2+b^2\right)^2 = \left(e^2+f^2\right)^2 - \left(c^2+d^2\right)^2$ or equivalently:

$$\left(a^2+b^2\right)^2 + \left(e^2+f^2\right)^2 - 2\left(c^2+d^2\right)^2 = 0 \tag{6}$$

Within the research made for this paper, no solution was found for any $a, b, c, d, e, f < 20000$. In the next sections, a set of distinct non zero integers that satisfy equation 5 will be called a seed, as a starting point for another strategy.

3 Polynomial parametrization

With bruteforce search being discarded, the idea is now to find a set of polynomials $a(X), b(X), c(X), d(X), e(X), f(X)$ that fulfill equation 5. Once this is done, the following polynomial is defined.

$$r(X) = \left(a(X)^2 + b(X)^2\right)^2 + \left(e(X)^2 + f(X)^2\right)^2 - 2\left(c(X)^2 + d(X)^2\right)^2 \quad (7)$$

This definition is obviously originating from equation 6 and a solution would be found with an integer root of $r(X)$ applied to all arithmetic progressions that define m. Luckily, finding a rational root (such event is easily checked using the rational root theorem) would be sufficient since all expressions may then be multiplied by the highest power of the denominator. For example, with $a(X) = \sum_{k=0}^{\eta_a} a_k X^K$ and $r(\frac{p}{q}) = 0$ this translates to:

$$q^{\eta_a} a(\frac{p}{q}) = \sum_{k=0}^{\eta_a} a_k p^k q^{\eta_a - k} \quad (8)$$

Of course, the elements derived that way must be in a correct order to solve the problem. There is a higher chance of obtaining the correct conditions when checking all possibilities between the permutations between the pairs (a, b), (c, d), (e, f) and mirroring all coefficients of all polynomials. The next parts of this paper are restricted to polynomials of degree 2 for which the constant terms $a_0, b_0, c_0, d_0, e_0, f_0$ are a given seed as defined above (distinct non zero elements). Starting from these numbers, there seems to be either an infinite number of polynomials satisfying equation 5, for example with $(1, 15, 2, 12, 4, 10)$, or no solution at all, for example with $(3, 88, 13, 55, 40, 51)$. As a showcase, the focus will be made on the seed $\sigma = (3, 77, 4, 70, 30, 44)$.

4 Description of the sweeping algorithm

At first the boundaries of any loop must be defined which was done using the extended euclidean algorithm in this work. Starting from the seed, the overall GCD is 1 (otherwise the integer solution are trivially derived from another seed) but each pair of numbers among the seed leads to a pair of Bézout coefficients. The maximal absolute value of any of these coefficients is g and any loop was chosen to span between $-g$ and $+g$. Note that a factor can be applied to restrict or extend the loops, however it seems to be a good value in general. Using the study case σ, $g = 26$ is obtained since we have $3 \times 26 - 77 \times 1 = 1$.

Then, it is noted that equation 5 exhibits two equal signs, thus it is convenient to start only with the left one, giving the first two pairs $a(X), b(X), c(X), d(X)$. The last pair $e(X), f(X)$ is searched later with the other member(s) already known.

The next step is to establish precisely which loops to cover. As the remaining equation uses polynomials of degree 2 up to a power of 4, the result is a polynomial of degree 8, thus giving 9 separate equations for each power of the variable, which must be fulfilled using 12 coefficients (3 per each of the 4 polynomials). 4 of those are already covered by the fact that a_0, b_0, c_0, d_0 are part of a seed i.e. $a_0 b_0(b_0^2 - a_0^2) = c_0 d_0(d_0^2 - c_0^2)$ for the coefficients of X^0 in equation 5. The equations for coefficients of X^7 and X^1 form a linear system in a_1, b_1, c_1, d_1:

$$\begin{cases} a_1(3a_2^2 b_2 - b_2^3) + b_1(a_2^3 - 3a_2 b_2^2) = c_1(3c_2^2 d_2 - d_2^3) + d_1(c_2^3 - 3c_2 d_2^2) \\ a_1(3a_0^2 b_0 - b_0^3) + b_1(a_0^3 - 3a_0 b_0^2) = c_1(3c_0^2 d_0 - d_0^3) + d_1(c_0^3 - 3c_0 d_0^2) \end{cases} \tag{9}$$

From there, let $\alpha = b_2(3a_2^2 - b_2^2)$, $\beta = a_2(a_2^2 - 3b_2^2)$, $\gamma = b_0(3a_0^2 - b_0^2)$ and $\delta = a_0(a_0^2 - 3b_0^2)$. In the hypothesis where c_1 and d_1 are known, the system 9 has a unique, well-defined solution for a_1 and b_1 when the determinant is non zero, in other words $\Delta = \alpha\delta - \beta\gamma \neq 0$. The most outer loop of the algorithm searches for a_2 and b_2 such that this condition is met, also fixing the value of $\tau = a_2 b_2(b_2^2 - a_2^2)$ contained in the coefficient of X^8 from the left side of equation 5. Each relevant pair of a_2 and b_2 was the starting point of a thread.

Then, the most outer loop of each thread was a search for c_2 and d_2 fulfilling $c_2 d_2(d_2^2 - c_2^2) = \tau$. From that point, let's consider the equations for coefficients of X^6 and X^2 in equality 5, without explicitely writing them here for simplicity. They both contain quadratic terms in a_1, b_1, c_1 and d_1. However, using the (actually existing) solutions derived from linear relationships 9, it is possible to write them in the following form:

$$\begin{cases} \lambda_\epsilon c_1^2 + \mu_\epsilon c_1 d_1 + \nu_\epsilon d_1^2 = \epsilon \\ \lambda_\phi c_1^2 + \mu_\phi c_1 d_1 + \nu_\phi d_1^2 = \phi \end{cases} \tag{10}$$

where none of the new variables depends on a_1 or b_1. Applying a cross product in order to combine both of the above, the terms in d_1^2 can be eliminated. After rearraging the result we obtain:

$$d_1 = \frac{\nu_\phi \epsilon - \nu_\epsilon \phi - c_1^2(\nu_\phi \lambda_\epsilon - \nu_\epsilon \lambda_\phi)}{c_1(\nu_\phi \mu_\epsilon - \nu_\epsilon \mu_\phi)} \tag{11}$$

Consequently, the next inner loop is on values of c_1. If $c_1 = 0$ or $\nu_\phi \mu_\epsilon - \nu_\epsilon \mu_\phi = 0$, then d_1 must be determined with another inner loop; otherwise the above formula is used to derive it. The values of c_1 and d_1 are used to find a_1 and b_1 with the system 9. The remaining equations are checked afterwards (i.e. coefficients of X^5, X^4 and X^3 in equality 5).

This way, the polynomials $a(X), b(X), c(X), d(X)$ are all defined. To get the other two polynomials a similar procedure as above can be applied, except that two polynomials and the entirety of their coefficients are fixed already. For example considering only the right equality of equation 5 and taking $c(X)$ and $d(X)$ as starting values. The values of e_0 and f_0 are once more fixed too because σ is a seed. A sweep on e_2 is performed, each value defining a thread; within each thread a single loop on f_2 checks that the determinant, defined similarly as above, is non zero and also that $e_2 f_2(f_2^2 - e_2^2) = \tau$ to satisfy the highest degree coefficient of equation 5. Again in a similar manner as before, the linear system allows to directly compute e_1 and f_1 from c_1 and d_1; the rest of the equations are then ckecked.

After all polynomials are defined, the new polynomial $r(X)$ can be computed as in equation 7 for all permutations of the pairs (in the end three of those are different and relevant). For each permutation, the rational root theorem is applied to try and find any rational root.

5 Specific example and near miss

One of the many (likely infinite) solutions starting from the chosen seed σ is the set of six polynomials:

$$\begin{cases} a(X) = 3X^2 + 9X - 12 \\ b(X) = 77X^2 - 29X + 2 \\ c(X) = 30X^2 - 45X + 15 \\ d(X) = 44X^2 + 7X - 1 \\ e(X) = 4X^2 + 17X + 4 \\ f(X) = 70X^2 - 55X + 10 \end{cases} \tag{12}$$

which was obtained after mirroring all of them and exchanging the last two pairs. The definition 7 gives $r(X) = 43341108X^8 - 103125264X^7 + 95670024X^6 - 55905648X^5 + 13684560X^4 + 4649952X^3 - 3846576X^2 + 911136X - 66792$ which can be simplified by 12 to get:

$$r(X) = 3611759X^8 - 8593772X^7 + 7972502X^6 - 4658804X^5 + 1140380X^4 + 387496X^3 - 320548X^2 + 75928X - 5566 \tag{13}$$

This polynomial has no rational root; however it has four real roots around -0.3866, 0.1345, 0.4484 and 1.2654. The second smallest can be approximated as the following continued fraction:

$$\rho = 0.13454810497520617607\ldots = \cfrac{1}{7 + \frac{1}{2+\cdots}} = [0, 1, 7, 1, 2, \ldots] \tag{14}$$

Changing the notation to $\rho = [[7, 2, \ldots]]$ we can extend it to $\rho = [[7, 2, 3, 5, 4, 1, 3, 1, 1121, 3, 1, 1, \ldots]]$. Then, stopping the series at the term 1121 gives a very good approximation, numerically $\rho \approx [[7, 2, 3, 5, 4, 1, 3, 1]]$ or:

$$\rho \approx \frac{p}{q} = \frac{923}{6860} \tag{15}$$

which is $\rho \approx 0.13454810495626822157\ldots$, very close to the actual root value. To obtain the near miss for the initial problem, we calculate the different elements of the structure using the model of equation 8 and the definition of the congrua's arithmetic progressions, leading to:

$$\begin{cases} q^4(e^2+f^2) = 122690194463810276 \\ q^4(c^2-d^2-2cd) = 230477545778066498 \\ q^4(a^2-b^2+2ab) = 229205527091232644 \\ q^4(c^2-d^2+2cd) = 278779976584780862 \\ q^4(a^2+b^2) = 200589096241058596 \\ q^4(e^2-f^2-2ef) = -52475667687057284 \\ q^4(a^2-b^2-2ab) = 167143044762381764 \\ q^4(e^2-f^2+2ef) = 165384618196290236 \\ q^4(c^2+d^2) = 255771552808841218 \end{cases} \tag{16}$$

There is a single negative number but it is also the smallest absolute value, quite convenently, so removing the sign keeps the solution valid. The only remaining common factor between the elements is 2 and the simplified version is then:

$$\begin{array}{ccc} 61345097231905138^2 & 115238772889033249^2 & 114602763545616322^2 \\ 139389988292390431^2 & 100294548120529298^2 & 26237833843528642^2 \\ 83571522381190882^2 & 82692309098145118^2 & 127885776404420609^2 \end{array} \tag{17}$$

Now, writing this with the same notation as structure 1 one can find that the set of equations 2 is almost satisfied with $S =$ 30176989143654525354729881691102729 except for the last with $G^2+E^2+C^2 =$ 30176989148103500825585759029118412. That means the sum of all lines, all columns and one diagonal adds up to the exact same number S, while the other diagonal sum differs by (relatively) less than a billionth, making it a very close near miss. It is possible to infinitely reduce that (relative) gap by going further in the continued fraction, however it looks like the common part of both diagonal sums will not exceed about 28% of their digits.

6 About polynomials with at least one rational root

The exact near miss shown above is not unique to seed σ, it was found across several other seeds within the scope of this paper. Actually, all seeds with $a_0, b_0, c_0, d_0, e_0, f_0 < 1000$ have been checked up to coefficients $\pm g/2$ as defined in section 4 and no actual solution was found, neither was a better approximation compared with section 5.

The work of *Hajdu et al.* [3] studies some properties of the coefficients of integer polynomials whose roots are exclusively rational. Nevertheless, this still gives hints on the properties of coefficients when at least one of the roots is rational, which is of interest in the present work. Considering a constant degree (for example $\eta_r = 8$ in section 4 onwards), there seems to be a minimum value for the height bound of the polynomial, i.e. the maximum absolute value among all coefficients. As a consequence, it could be useful to use seeds containing bigger values, which are likely to lead to bigger coefficients of $r(X)$ in equation 7. This might be related to the number of factors, since more values are checked using the rational root theorem in that case.

The other aspect mentioned in [3] is the factors of the coefficients. Once again in the hypothesis of a given degree, it seems more probable to have at least one rational root when all coefficients are divisible by 2 or 3. By extension, it may be best to use seeds where all elements are divisible by 2 or 3, such as $(36, 456, 38, 448, 369, 399)$. However, no seed featuring this property has led to a proper polynomial parametrization within the scope of this work (see above). Similarly to the height bound indicated above, it looks like we can extend this divisibility property to a given set of prime numbers, potentially as populated as possible, thus much more challenging in the determination of $a(X), b(X), c(X), d(X), e(X), f(X)$.

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Conclusion

Given the initial problem of the magic square of squares, the intent of this paper was not only to propose and describe a searching method, but also to illustrate it and demonstrate its capabilities. Presumably, using community effort and computing power, a solution is bound to emerge.

References

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